

FILTER DESIGN TEST CASE

Low-Pass Filter - Band 8 (900 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Low-Pass Filter filter targeting Band 8 (900 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 3867.2 MHz and bandwidth of 100.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Low-Pass Filter
- Frequency Band: Band 8 (900 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: 36° YX LiTaO3
- Package: CSP (Chip Scale Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the Low-Pass Filter design targeting Band 8 (900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3867.2 MHz
Bandwidth (3 dB):	100.4 MHz
Insertion Loss Target:	≤ 1.3 dB
Return Loss Target:	≥ 11.0 dB
Out-of-Band Rejection:	≥ 51.0 dB
Isolation (Duplexer):	≥ 24.0 dB
Q Factor:	9176
Temperature Coefficient:	-16.6 ppm/°C
Die Size:	1.1 x 1.7 mm
Wafer Size:	8-inch
Number of Resonators:	8
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Low-Pass Filter template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 3867.2 MHz, BW 100.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11602 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 11.0 dB
- Out-of-band rejection at +/- 151 MHz offset >= 51.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 8.0 MHz
- Power handling shall be >= +26 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2024-07-06
Devices Tested: 52
Insertion Loss (meas): 1.04 dB
Return Loss (meas): 13.4 dB
Rejection (meas): 54.0 dB
Isolation (meas): 28.5 dB
Q Factor (meas): 9176
Group Delay Variation: 5.8 ns
Power Handling: +31 dBm
Die Yield: 93.4%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Low-Pass Filter design demonstrated acceptable performance for Band 8 (900 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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Approver:	Dr. Hans Weber	Date:	2024-07-10

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